METRIC

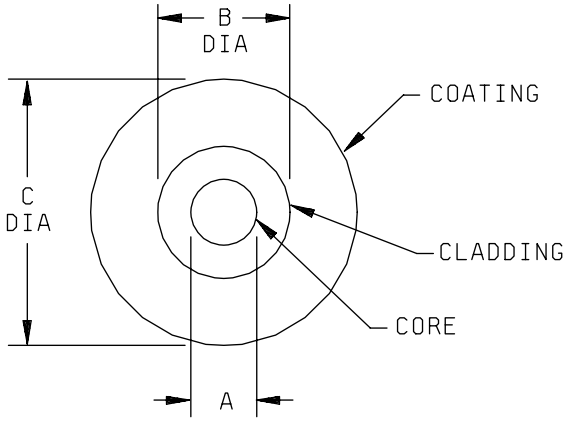
MIL-PRF-49291/1B
18 December 2002
SUPERSEDING
MIL-F-49291/1A
8 July 1992

PERFORMANCE SPECIFICATION SHEET

FIBER, OPTICAL, TYPE I, CLASS I, SIZE III, COMPOSITION A.
WAVELENGTH B, RADIATION RESISTANT (METRIC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-PRF-49291.



Fiber	PIN	Dimensions		
		A (µm)	B (µm)	C (µm)
Standard	M49291/1-01	50.0 ± 3	125 ± 1	245 ± 10
Standard	M49291/1-02	50.0 ± 3	125 ± 1	500 ± 25

FIGURE 1. Dimensions and configuration of optical fiber construction.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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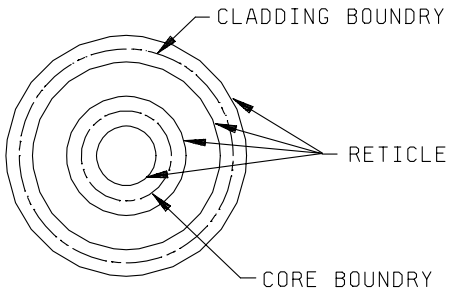

Circle (solid)	Diameter (μm)
Inner	47.0
Second	53.0
Third	124.0
Fourth	126.0

FIGURE 2. Tolerance fields.

DIMENSIONS AND CONFIGURATION:

Diameter: See figures 1 and 2. (Diameter requirements are based on selecting fibers with end-points tolerances of 125 ± 1 μm from production with a 125 ± 2 μm tolerance.)

Ovality:
Core: ≤ 6 percent.
Cladding: ≤ 2 percent.

Offset:
Core-to-cladding: ≤ 1.5 μm.
Fiber-to-coating: coating-cladding concentricity error ≤ 12.5 μm
(overall coating concentricity ratio (OCCR) ≥ 0.70 for 250 μm diameter coatings and ≥ 0.84 for 500 μm diameter coatings).

Splices: Not allowed.

Fiber mass/unit length: 0.1 kg/km maximum.

Tensile proof: 690 MPa.

Change in optical transmittance: Measurements to be made at 1300 ± 20 nm.

Maximum attenuation rate: 3.5 dB/km at 850 ± 20 nm
1.0 dB/km at 1300 ± 20 nm

Numerical aperture: 0.20 ± 0.015 at 850 nm ± 25 nm.

Bandwidth: ≥ 500 MHZ-km at 1300 nm ± 20 nm.
≥ 500 MHZ-km at 850 nm ± 20 nm.

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Macrobend attenuation: Performed at 1300 ± 20 nm.

Chromatic dispersion: The zero dispersion wavelength shall be not less than 1295 nm and not greater than 1330 nm. The dispersion slope at the zero dispersion wavelength shall not be greater than 0.11 ps/nm^{2}km.

Transient Attenuation: ≤ 1.5 dB at 1300 nm.

Light launch conditions: In accordance with EIA-455-50.
Wavelength: 1300 ± 25 nm.
Source type: LED with FWHM spectral width ≤ 160 nm.

ENVIRONMENTAL:

Temperature range:
Operating: -55° C to +85°
Nonoperating: -62° C to +85° C
Storage: -62° C to +85° C

Fluid immersion aging: Not applicable.

Nuclear radiation resistance: The nuclear radiation resistance characteristics of this optical fiber are classified and are contained in an appendix to this specification. Application to receive this appendix must be made through the US Army Communications- Electronics Command, ATTN: AMSEL-LC-LEO-E-EP, Fort Monmouth, NJ 07703-5023. Information concerning security clearance classification and "need to know" must be detailed in the request.)

Fungus: Applicable.

Dynamic tensile strength: Applicable.

QUALITY CONFORMANCE:

In group A testing length may be measured using mechanical methods.

In group C testing the mechanical strippability test may be omitted if the optical fiber coatings have not changed from when the mechanical strippability test was last performed. The manufacturer shall provide a certificate of compliance for mechanical strippability in the group C test report.

Part or identifying number (PIN). (See figure 1 and table I):

M49291/1-01
M49291/1-02

TABLE I. Supersession data.

PIN	Superseding
M49291/1-01	D49291/1-01
M49291/1-02	None

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Custodians:
Army - CR
Navy - SH
Air Force - 11
NASA - NA
DLA - CC

Review activities:
Navy - AS
Air Force - 13, 19, 93, 99
DIA - DI

Preparing activity:
DLA - CC

(Project 6010-0048)

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